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Prevalence and risk factors of frailty among people in rural areas: a systematic review and meta-analysis

Rui Xu,[1] Qiufang Li,[1] Feifei Guo,[1] Maoni Zhao,[2] Luyao Zhang[1]

ABSTRACT

Objective Rural elderly is possible more frail due to the limited medical resources and lower socioeconomic status. Given the negative healthy outcomes caused by frailty, knowing the epidemiology of frailty in rural areas is of great importance. However, the prevalence of frailty in rural elderly varies widely. We tried to synthesize the existing evidences for the prevalence and risk factors of frailty in rural areas.

Design A systematic review and meta-analysis.

Data sources PubMed, Embase, MEDLINE, Cochrane Library, Web of Science and Scopus were used to identify the articles from inception to 30 April, 2019.

Eligibility criteria Observational studies providing cross-sectional data on the prevalence of frailty were extracted.

Data synthesis The pool prevalence of frailty was calculated by the random effects model and the odds ratio (OR) and 95% CI were used to calculated the risk factors.

Results The literature search yielded 2210 articles, of which 25 met the study criteria and were included in this analysis. The pooled prevalence of frailty and prefrailty were 18% (95% CI 15% to 21%, I^2 = 98.4%, p = 0.000) and 50% (95% CI 45% to 56%, I^2 = 98.4%, p = 0.000), respectively. Age (OR=1.05, 95%CI 1.03 to 1.08), cognitive impairment (OR=1.97, 95%CI 1.41 to 2.54), depressive symptom (OR=1.24, 95%CI 1.14 to 1.34), risk of malnutrition (OR=2.49, 95%CI 1.51 to 3.48), ADL disability (OR=2.59, 95%CI 1.71 to 3.48), poor self-perception of health (OR=2.42, 95%CI 1.39 to 3.45) were associated with frailty.

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Conclusions Frailty influences almost one in five elderly in rural areas. Increasing age, cognitive impairment, depressive symptom, risk of malnutrition, ADL disability and poor self-perception of health were all risk factors for frailty. It is necessary to take strategies to prevent and improve the frailty of the elderly in rural areas.

Strengths and limitations of this study

(1) In this study, we conducted a comprehensive systematic literature search.

(2) This study focused on the frailty of elderly in rural areas that are easily neglected.

(3) The main limitation is the heterogeneity among studies of this meta-analysis.

INTRODUCTION

An aging population is a common phenomenon experienced by all countries in the world,[1] which brings considerable challenges for the planning and delivery of healthcare services internationally. The most problematic manifestation of population aging is frailty.[2] Frailty is an age-related medical syndrome, which is characterized by diminished strength, endurance, and reduced physiologic function that increases an individual's vulnerability for developing increased dependency and/or death.[3] Prefrailty is an intermediate state between frailty and robustness, which is a transitional and potentially reversible risk state before the onset of frailty.[4] Frailty is associated with negative healthy outcomes, such as hospitalization, institutionalization/dependency and premature mortalities.[5]

With the continuous expansion of the study population, more and more researches begin to pay attention to the frail elderly in rural areas. From high-income countries to sub Saharan Africa, under the guidance of city oriented development path, the gap between urban and rural areas continues to expand.[6] The elderly person in the rural context, in addition to the peculiarities of the aging process,[7] shows a series of environment related characteristics, such as limited educational resources, limited accessibility of healthcare services and resources, lower socioeconomic status, and all of these potential factors may cause the emergence or advancement of frailty.[8 9] Many countries have explored the prevalence of frailty in rural elderly and the results are quite different, for example 43.4% in Brazil,[7] 15.2% in Sri Lanka,[10] 38.8%in India,[11]

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18.8% in French[12] and even in the same country, the prevalence of frailty varies greatly due to differences in assessment tools, research environments and so on. Hence, it's necessary to synthesize the existing body of evidence and offer a perspective on the prevalence of frailty in rural areas, which are critically important to developing strategies and disease prevention.

The objectives of this systematic review were to conduct a meta-analysis synthesizing the pooled prevalence and risk factors of frailty in rural areas.

METHODS

Protocol

This review was conducted according to the MOOSE statement which was for the meta-analysis of observational studies in epidemiology .[13]

Search strategy

We conducted a comprehensive search of the literature using six electronic databases. PubMed, Embase, MEDLINE, Cochrane Library, Web of Science and Scopus were searched from their dates of inception through 30 April 2020 with languages restricted to English. A combination of title/keyword/abstract and Medical Subject Headings (Mesh) were used and the following key words were used: "frailty", "frail", "frail*", "rural", "village", "countryside" and "rural*". Reference lists of relevant literatures were hand-searched for addition studies. The full electronic search strategy is shown in table 1.

Table 1 Search strategy for PubMed, Embase, Web of Science, Cochrane, Medline and Scopus

Database	Search strategy
PubMed	#1：(((frailty[Title/Abstract]) OR (frail[Title/Abstract])) OR (frail*[Title/Abstract])) OR (frailty[MeSH Terms]) #2：(((rural[Title/Abstract]) OR (village[Title/Abstract]))OR (countryside[Title/Abstract])) OR (rural*[Title/Abstract]) #3：#1 AND #2
Embase	#1：(frailty:ab,ti OR frail:ab,ti OR frail*:ab,ti) #2：(rural:ab,ti OR countryside:ab,ti OR village:ab,ti OR rural*:ab,ti) #3：#1 AND #2
Web of Science	#1:TI=(frailty OR frail OR frail*) OR TS=(frailty OR frail OR frail*) #2：TI=(rural OR village OR countryside OR rural*) OR TS=(rural OR village OR countryside OR rural*)

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	#3：#1 AND #2

Table 1 Continued

Database	Search strategy
Cochrane	#1：(frailty):ti,ab,kw OR (frail):ti,ab,kw OR (frail*):ti,ab,kw #2：(rural):ti,ab,kw OR (village):ti,ab,kw OR (countryside):ti,ab,kw OR (rural*):ti,ab,kw #3：#1 AND #2
Medline	#1：AB frailty OR AB frail OR AB frail* #2：AB rural OR AB village OR AB countryside OR AB rural* #3：#1 AND #2
Scopus	#1:(TITLE-ABS-KEY (frailty) OR TITLE-ABS-KEY (frail) OR TITLE-ABS-KEY (frail*)) #2：(TITLE-ABS-KEY (rural) OR TITLE-ABS-KEY (village) OR TITLE-ABS-KEY (rural*) OR TITLE-ABS-KEY (countryside)) #3：#1 AND #2

Inclusion and exclusion criteria

The inclusion criteria were as follows: study designs were observational providing cross-sectional data on the prevalence of frailty; participants were over 60 years of age; sample or subsample consisted of people in rural areas. The exclusion criteria were as follows: participants were restricted to specific population such as patients with cancer, kidney disease, etc.; review, editorial, comment, and conference abstract were excluded.

Data extraction

Two investigators extracted data independently and disagreements were resolved by discussion. Data were recorded in the Microsoft Excel file format from each eligible study including the following variables: first author, year and country of publication, sample size, assessment tools of frailty, prevalence of frailty and prefrailty, and risk factors mentioned.

Methodological quality assessment

All eligible articles were subjected to a quality assessment by two investigators independently through the criterion proposed by Loney et al[14]which was designed to critically assess studies about the prevalence. A third author resolved quality disagreements.

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Statistical analysis

Heterogeneity among studies was tested using Chi-square test and the I^2 statistic. Studies were considered to have significant heterogeneity if the *p* value was less than 0.1. The I^2 statistic was used to assess the degree of heterogeneity, with I^2 values of 25%, 50%, and 75% being considered to indicate low, moderate, and high heterogeneity, respectively. Random-effects model was used to calculate the pooled prevalence of frailty if significant heterogeneity was present; otherwise a fixed-effects model was used. We tried to explore the potential reasons for heterogeneity by meta-regression analysis which can associate difference in effect sizes with study characteristics. Factors were examined were: year of publication, study size (comparing studies of over 1000 samples with smaller studies, and comparing sample size as continuous variable), assessment tool (comparing studies using the Fried Phenotype including the Modified Fried Phenotype and studies using other tools), country, level of development. The prevalence of frailty and prefrailty were extracted from all included studies in order to calculate the pooled prevalence and the odds ratios (ORs) and associated 95% CIs were extracted to assess the risk factors of frailty on rural elderly. Begg's test and Egger's test were used to assess publication bias.

Statistical analyses were conducted using Stata SE, version 12.0 (Stata Corp College Station, TX), and a p value of less than 0.05 was considered significant.

Patient and public involvement

There was no direct patient or public involvement in this study.

RESULTS

Study process

The systematic search of the six databases retrieved 2210 articles. After 1336 duplicate studies were excluded, a further 783 studies were excluded through screening the title and abstract. Of the 91 studies left, 63 were excluded from the full-text review due to non-data-based article (e.g. editorial, commentary, review, conference abstract) (n=15), same sample (n=11), non-English (n=7), no date of frailty in rural elderly (n=25), sample from a specific population (e.g. patients with cancer , disabled individuals) (n=5), one more form of frailty assessment (n=3).

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Finally, 25 published articles,[7-12 15-33] involving a total of 32848 individuals in rural areas, were seemed eligible and included in the meta-analysis (Fig. 1). The characteristics of studies were showed in Table 2. One of these studies included data from four countries.[33]

Fig 1. Flow chart of systematic review

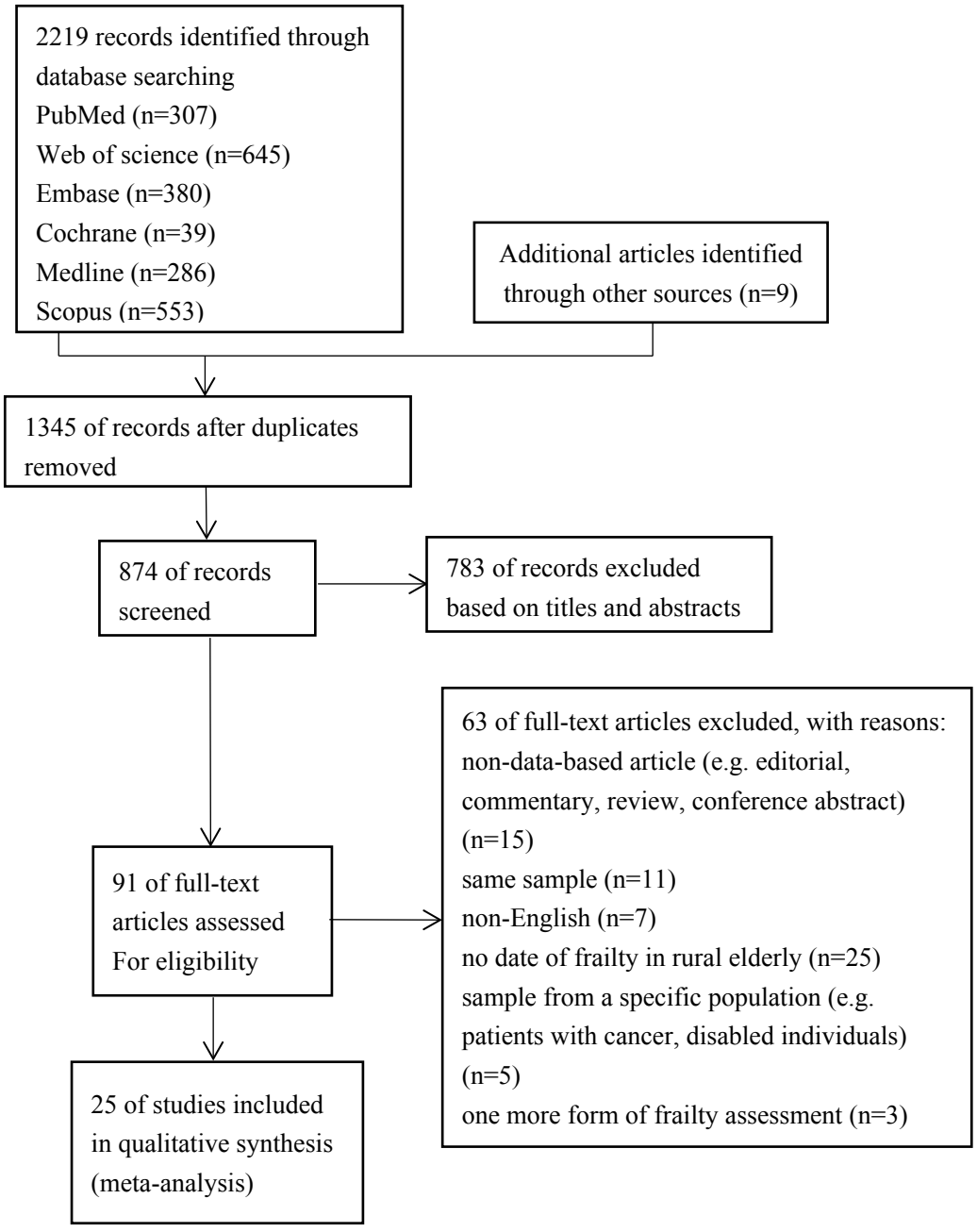


Table 2 Characteristics of studies

First author(year)	Country	Sample size	Frailty assessment	Prevalence (%)		Risk factors assessed	Quality Score
				frail	prefrail		
Llano2019(7)	Brazil	820	FP	43.4	37.1	income; education level; nutritional status，physical inactivity ,cognitive deficit ,poor self-perceived health	6
Manrique-Espinoza2016(8)	Mexico	558	FP	8.6	52.9	ADL and IADL disability, depression symptom, educational level, self-reported health	6
Gu2019(9)	China	4323	FP	6.8	49.4	age, stroke history, vision decrease, anemia	6
Siriwardhana2019(10)	Sri Lanka	746	FP	15.2	48.5	age, longest-held occupation,	4
Dasgupta2019(11)	India	165	TFI	38.8	—	educational level，2 chronic diseases	4
Avila-Funes2017(12)	French	176	FP	18.8	—	—	3
Huang2020(15)	China	1014	FP	17.6	59.4	depressive symptoms; urinary incontinence; abnormal performance of TUG; malnutrition	5
Nguyen2019(16)	Vietnam	512	FP	21.7	65.6	age, marital status, occupation, comorbidities, cognitive impairment, history of fall in the last 12 months	5
Lewis2018(17)	Tanzania	196	the Modified FP	9.25	—	—	5
Ahmad2018(18)	Malaysia	2324	FP	9.4	57.9	—	6

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Yoon2018(19)	Korea	104	FP	16.5	49.6	—	3

Table 2 Continued

First author(year)	Country	Sample size	Frailty assessment	Prevalence (%)		Risk factors assessed	Quality Score
				frail	prefrail		
Li2018(20)	China	3048	FI	15.8	67.8	self-neglect, age, sex and alcohol drinking	5
Del,Brutto2016(21)	Ecuador	311	EFS	31	23	—	5
Dent2016(22)	Australia	1501	FI	25	—	health service use	6
Kim2019(23)	Korea	808	KYC	35.5	—	perceived neighborhood walkability ,environmental pollution	6
Boulos2016(24)	Lebanon	1120	SOF	36.4	30.4	marital status, education level, Physical health status, Mental health status	6
Zhu2016(25)	China	1478	FP	12.0	42.9	high sensitivity C-reactive protein	6
Cakmur2015(26)	Turkey	168	FP	7.1	47.3	—	4
Curcio2014(27)	Colombia	1878	FP	12.2	53	age, gender, health status variables , functional covariate variables, psychosocial variables	4
Ocampo-Chaparro2019(28)	Colombia	688	FP	20.6	64.4	—	6
Wu2017(29)	China	3048	FP	8.1	53.6	—	8
Abe2019(30)	Japan	1576	KYC	20.8	—		8
Ma2018(31)	China	2353	FI	12.9	—	—	7
Jung2016(32)	Korea	380	FP	17.4	52.6	—	4
Llibre2018(33)	Peru; Mexico; China; India	552; 1000; 1002; 999	the Modified FP	16.8; 15.7; 5.4; 15.4	—	—	7

Note: ADL=activity of daily living; IADL=instrumental activity of daily living; TUG=timed up-and-go; FP=the Fried phenotype; FI=Frailty Index; TFI=Tilburg Frailty Indicator; EFS=the Edmonton Frail Scale; SOF=the Study of Osteoporotic Fractures index; KYC=The Kaigo-Yobo checklist.

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Prevalence of frailty and prefrailty

The prevalence of frailty and prefrailty in included studies range from 5.4%[33] to 43.4%[7], and from 23%[21] to 67.8%[20], respectively. Meta-analysis showed the pooled prevalence of frailty 18% (95% CI 15% to 21%, I^2 = 98.4%, p = 0.000) in rural elderly, with the pooled prevalence of prefrailty of 50% (95% CI 45% to 56%, I^2 = 98.4%, p = 0.000) based on a random-effects model (Fig. 2). Heterogeneity statistics (Chi-square test=1651.51; p=0.000; I^2=98.4%) indicated significant variability among studies. According to the results of Begg'test (Pr=0.650) and Egger's test (p=0.061), there was no evidence of publication bias.

Meta-regression

Factors were examined including year of publication, sample size (continuous), sample size ($>$1000 vs $\leq$1000), assessment tool (the Fried Phenotype(the Modified Fried Phenotype were included) vs other), country (Asia vs South and North America vs Africa vs other), level of development (developing country vs developed country) in meta-regression analyses. Assessment tool showed significant relationship to the prevalence of frailty in rural elderly (β=0.12±0.04, t=3.24, p=0.003) while the rest of factors were not significant (Table 3). Since assessment tool was the factor that is related to the prevalence of frailty in meta-regression analyses, we conducted a subgroup analysis according to the type of assessment tools. The pooled estimates of frailty prevalence based on the Fried Phenotype and other tools were 15% (95% CI 12%-17%; z = 10.85; p = 0.000) and 18% (95% CI 15%-21%; z = 8.56; p = 0.000) respectively.

Risk factors

We identified six potential risk factors including age, cognitive impairment, depressive symptom, risk of malnutrition, ADL disability, poor self-perception of health, which were associated with frailty in rural elderly for the pooled analysis (Table 4).

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Table 3 Univariate metaregression for prevalence of frailty in rural elderly

Factors	Meta-regression coefficient	95%CI	P-value
Year of publication	0.01	-0.01 to 0.04	0.316
Sample size, continuous	-0.00	-0.00 to 4.00e	0.078
Sample size (＞1000 vs ≤1000)	0.05	-0.02 to 0.13	0.163
Assessment tool (the Fried Phenotype vs other)	0.12	0.04-0.19	0.003
Country (Asia vs South and North America vs Africa vs other)	0.01	-0.03 to 0.06	0.608
Level of development (developing vs developed country)	0.05	-0.04 to 0.15	0.263

Table 4 Pooled risk factors of frailty

No.	Risk factors	OR	95%CI	I^2	P-value
1	Age	1.05	1.03-1.08	0.0%	0.000
2	Cognitive impairment	1.97	1.41-2.54	0.0%	0.000
3	Depressive symptom	1.24	1.14-1.34	68.5%	0.000
4	Risk of malnutrition	2.49	1.51-3.48	62.0%	0.000
5	ADL disability	2.59	1.71-3.48	0.0%	0.000
6	Poor self-perception of health	2.42	1.39-3.45	0.0%	0.000

DISCUSSION

Based on 25 studies involving a total of 32848 participants living in rural areas, the estimated pooled prevalence of frailty and prefrailty in rural elderly are 18%, and 50%, respectively. When assessing potential risk factors associated with frailty in rural elderly, six factors including increasing age, cognitive impairment, depressive

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symptom, risk of malnutrition, ADL disability and poor self-perception of health were associated with frailty.

The pooled prevalence of frailty in rural elderly from this meta-analysis was 18%, almost twice which of the global estimate.[34] Frailty was more common among rural areas, which may be explained by these reasons: (1) lower socioeconomic status among residents living in rural communities, (2) limited accessibility of healthcare services and resources, (3) relatively unhealthy lifestyle and limited health care awareness.[15 16] The higher prevalence of frailty in the rural areas in this study may be multifactorial, which cannot be simply explained by the rurality. The higher prevalence of frailty in rural areas caused more healthy problems and burden. Studies from Canadian[35], China[36] and Korea[37] found that frail rural elderly have higher mortality than that in the city or town. Hence, it's time to focus and improve the health care in rural areas, expect for improving the distribution of medical resources, strategies including support services, nutritional supplementation, exercise interventions and health education may also be useful to prevent and delay the occurrence and development of frailty.[32] Half of the rural elderly in this meta-analysis were prefrailty, indicating the urgency to establish and implement early frailty prevention strategies among rural areas.

The potential risk factors associated with frailty estimated in rural elderly were increasing age, cognitive impairment, and depressive symptom, risk of malnutrition, ADL disability and poor self-perception of health. Aging can be conceptualized as a deficit accumulation process of each physiological system, taking place in different individuals in different ways, with a variety of rates for different organ systems, making it necessary to find out prevalence and associated factors of different age groups of the elderly in rural areas.[38] Limited data made it difficult to conduct a subgroup of age in this meta-analysis, but evidence for rural elderly in Sri Lanka had found the prevalence of frailty has almost doubled with the increasing of age (group: 75-79 years vs ≥80 years; RR: 4.07 vs 7.02).[10] Park[39] found factors affecting frailty by age groups was quite different by analyzing the data of 22868 elderly people, telling us to implement targeted interventions of frailty prevention, which can reduce the

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pressure in rural areas with limited medical resources. Cognitive impairment was associated with frailty in rural areas, consistent with results among the world that poor cognitive function is strongly closed associated with pre-frailty and frailty subgroups in older populations,[40] accurate evaluation of cognitive function and encouraging an active life style maybe the useful services aiming to prevent the premature health decline effects in rural elderly. Frailty and depression share the same pathophysiological mechanism and individuals with depressive symptoms usually have less physical activity, which will increase the probability of frailty.[41 42] Weight loss, reduced intake of calorie and specific dietary pattern are associated with changes in body composition and physical function, which also increase the risk of frailty.[43 44] Cognitive impairment, depressive symptom and risk of malnutrition/malnutrition were always seemed associated with frailty. , and also reminds us the mental health and nutrition improvement should be included in the frailty intervention programs for rural elderly.[15] ADL disability means the decreased ability of self-care, affecting eating ability and physical activity, further causing malnutrition and muscle decline, eventually leading to frailty.[45] Self-perception of health has been identified as a highly sensitive variable in elderly health estimations and a valid predictor of morbidity[8], and Jylha[46] found self-appraisal has biological and cognitive underpinnings, existing as a predictor of frailty. Actually, rural older persons are exposed to a greater inequitable and precarious socioeconomic and health conditions throughout their life courses, and as a result, perception of the elderly regarding adverse experiences may predispose them to frailty.[8]

In this meta-regression analysis, we found the assessment tool of frailty perhaps related to the prevalence of frailty in rural areas while the years of publication, sample size, sample size, country and level of development were not the influence factors. The different prevalence of frailty among assessment methods tell us that the importance of finding suitable tools for diagnosing frailty of rural areas in different countries. The reason for heterogeneity among articles is still not clear, and more methods should be used to try to find out the origin of heterogeneity.

The strength of this study is to focus on the frailty of elderly in rural areas that are

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easily neglected. However, there are still several limitations. The main limitation is the heterogeneity among studies. Moreover, the included articles mainly from Asia, South and North America, lacking studies from Europe and Africa, perhaps due to the difference in number and distribution of rural population. Third, the quality of the studies included was moderate, and we should be careful with research results.

CONCLUSION

The pooled prevalence of frailty of elderly was 18% in rural areas and many factors were associated with frailty, including age, cognitive impairment, depressive symptom, risk of malnutrition, ADL disability and poor self-perception of health. As a relatively neglected elderly group, evidence for the prevalence of frailty and its risk factors of rural elderly are important to the government to formulate policies related to aging.

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Author Contributors QL and MZ conceived and designed the study strategy; RX and FG completed the work of the article search, article assessment, data extraction, quality assessment and data analysis independently; LZ participated in the discussion when there was disagreement; RX wrote the manuscript. The definitive manuscript was read and approved by all authors.

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For peer review only

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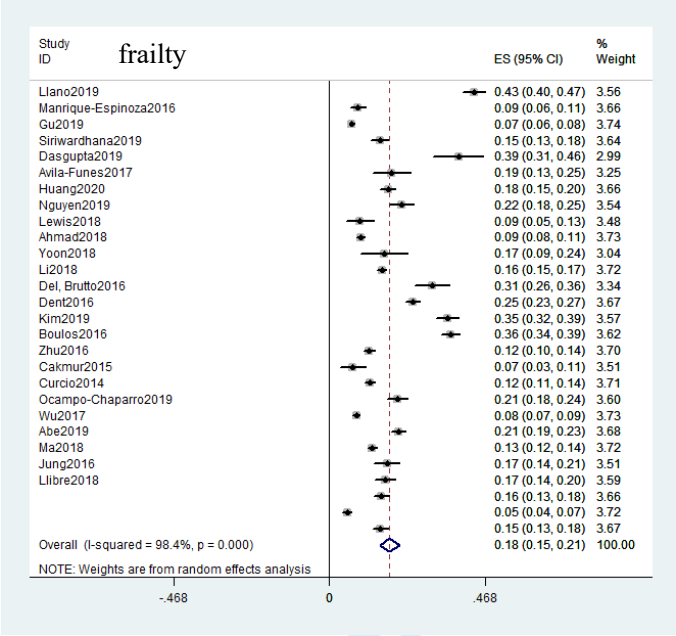
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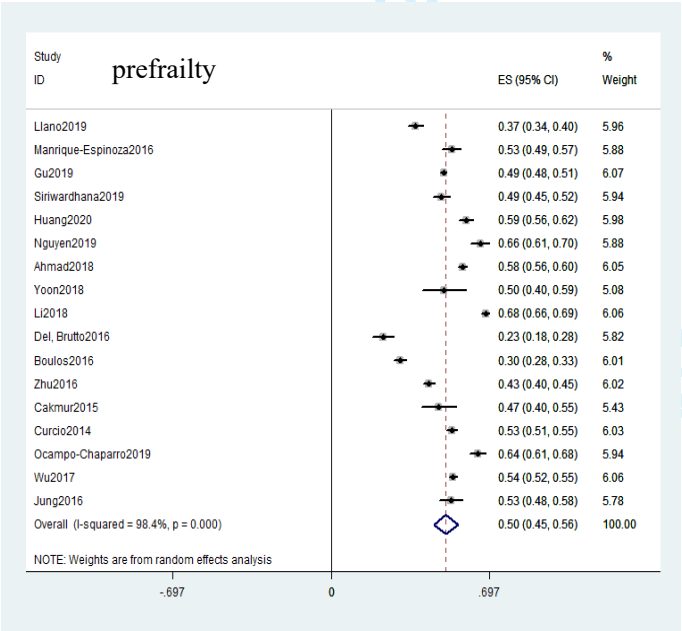
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Study ID
frailty
ES (95% CI)
% Weight
Llano2019 0.43 (0.40, 0.47) 3.56
Manrique-Espinoza2016 0.09 (0.06, 0.11) 3.66
Gu2019 0.07 (0.06, 0.08) 3.74
Siriwardhana2019 0.15 (0.13, 0.18) 3.64
Dasgupta2019 0.39 (0.31, 0.46) 2.99
Avila-Funes2017 0.19 (0.13, 0.25) 3.25
Huang2020 0.18 (0.15, 0.20) 3.66
Nguyen2019 0.22 (0.18, 0.25) 3.54
Lewis2018 0.09 (0.05, 0.13) 3.48
Ahmad2018 0.09 (0.08, 0.11) 3.73
Yoon2018 0.17 (0.09, 0.24) 3.04
Li2018 0.16 (0.15, 0.17) 3.72
Del, Brutto2016 0.31 (0.26, 0.36) 3.34
Dent2016 0.25 (0.23, 0.27) 3.67
Kim2019 0.35 (0.32, 0.39) 3.57
Boulos2016 0.36 (0.34, 0.39) 3.62
Zhu2016 0.12 (0.10, 0.14) 3.70
Cakmur2015 0.07 (0.03, 0.11) 3.51
Curcio2014 0.12 (0.11, 0.14) 3.71
Ocampo-Chaparro2019 0.21 (0.18, 0.24) 3.60
Wu2017 0.08 (0.07, 0.09) 3.73
Abe2019 0.21 (0.19, 0.23) 3.68
Ma2018 0.13 (0.12, 0.14) 3.72
Jung2016 0.17 (0.14, 0.21) 3.51
Llibre2018 0.17 (0.14, 0.20) 3.59
0.16 (0.13, 0.18) 3.66
0.05 (0.04, 0.07) 3.72
0.15 (0.13, 0.18) 3.67
Overall (I-squared = 98.4%, p = 0.000) 0.18 (0.15, 0.21) 100.00
NOTE: Weights are from random effects analysis
-.468
0
.468


Study ID
prefrailty
ES (95% CI)
% Weight
Llano2019 0.37 (0.34, 0.40) 5.96
Manrique-Espinoza2016 0.53 (0.49, 0.57) 5.88
Gu2019 0.49 (0.48, 0.51) 6.07
Siriwardhana2019 0.49 (0.45, 0.52) 5.94
Huang2020 0.59 (0.56, 0.62) 5.98
Nguyen2019 0.66 (0.61, 0.70) 5.88
Ahmad2018 0.58 (0.56, 0.60) 6.05
Yoon2018 0.50 (0.40, 0.59) 5.08
Li2018 0.68 (0.66, 0.69) 6.06
Del, Brutto2016 0.23 (0.18, 0.28) 5.82
Boulos2016 0.30 (0.28, 0.33) 6.01
Zhu2016 0.43 (0.40, 0.45) 6.02
Cakmur2015 0.47 (0.40, 0.55) 5.43
Curcio2014 0.53 (0.51, 0.55) 6.03
Ocampo-Chaparro2019 0.64 (0.61, 0.68) 5.94
Wu2017 0.54 (0.52, 0.55) 6.06
Jung2016 0.53 (0.48, 0.58) 5.78
Overall (I-squared = 98.4%, p = 0.000) 0.50 (0.45, 0.56) 100.00
NOTE: Weights are from random effects analysis
-.697
0
.697

MOOSE Checklist

Section	Item No.	Checklist Item
Background		
Problem definition	1	Synthesize the existing evidences for the prevalence and risk factors of frailty in rural areas.
Hypothesis statement	2	The elderly in rural areas has a high prevalence of frailty.
Description of study outcome(s)	3	The pooled prevalence of frailty and prefrailty were 18% and 50%, respectively. Age, cognitive impairment, depressive symptom, risk of malnutrition, ADL disability and poor self-perception of health were associated with frailty.
Type of exposure or intervention used	4	The prevalence of frailty.
Type of study designs used	5	Study designs were observational providing cross-sectional
Study population	6	Elderly in rural areas.
Search strategy		
Qualifications of searchers (eg, librarians and investigators)	7	This search was conducted by two postgraduates with relative training.
Search strategy , including time period included in the synthesis and keywords	8	It was shown in table 1.
Effort to include all available studies , including contact with authors	9	Reference lists of relevant literatures were hand-searched for addition studies in order to obtain all available studies.
Databases and registries searched	10	PubMed, Embase, Web of Science, Cochrane, Medline and Scopus were used to search.
Search software used, name and version , including special features used (eg, explosion)	11	It was EndNote x9.
Use of hand searching (eg, reference lists of obtained articles)	12	There were 9 articles through the hand searching.
List of citations located and those excluded, including justification	13	It was shown in Fig 1.
Method of addressing articles published in languages other than English	14	Not included.

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Continued

Section	Item No.	Checklist Item
Method of handling abstracts and unpublished studies	15	Not included.
Description of any contact with authors	16	There was no any connection with authors.
Methods		
Description of relevance or appropriateness of studies assembled for assessing the hypothesis to be tested	17	Studies included assembled for assessing the hypothesis to be tested.
Rationale for the selection and coding of data (eg, sound clinical principles or convenience)	18	Data were recorded including the following variables: first author, year and country of publication, sample size, assessment tools of frailty, prevalence of frailty and prefrailty, and risk factors mentioned.
Documentation of how data were classified and coded (eg, multiple raters, blinding, and interrater reliability)	19	All eligible articles were subjected to a quality assessment by two investigators independently.
Assessment of confounding (eg, comparability of cases and controls in studies where appropriate)	20	Not included.
Assessment of study quality, including blinding of quality assessors; stratification or regression on possible predictors of study results	21	The blinding was used for the assessment of study quality.
Assessment of heterogeneity	22	Heterogeneity statistics (Chi-square test=1651.51; p=0.000; I2=98.4%) indicated significant variability among studies.
Description of statistical methods (eg, complete description of fixed or random effects models, justification of whether the chosen models account for predictors of study results, dose-response models, or cumulative meta-analysis) in sufficient detail to be replicated	23	A random-effects model was used according to the results of heterogeneity (I^2=98.4%)

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Continued

Section	Item No.	Checklist Item
Provision of appropriate tables and graphics	24	Tables and graphics were shown in this article.
Results		
Graphic summarizing individual study estimates and overall estimate	25	It was shown in Fig 2.
Table giving descriptive information for each study included	26	It was shown in Table 2.
Results of sensitivity testing (eg, subgroup analysis)	27	It was shown in Table 3
Indication of statistical uncertainty of findings	28	Including the 95% CI and P-value.
Discussion		
Quantitative assessment of bias (eg, publication bias)	29	There was no evidence of publication bias according to the results of Begg' s test (Pr=0.650) and Egger ' s test (p=0.061).
Justification for exclusion (eg, exclusion of non- English-language citations)	30	In order to obtain the complete data and universal results.
Assessment of quality of included studies	31	The quality of included studies was in the level of moderate.
Conclusions		
Consideration of alternative explanations for observed results	32	The quality of original studies.
Generalization of the conclusions (ie, appropriate for the data presented and within the domain of the literature review)	33	The pooled prevalence of frailty of elderly was 18% in rural areas and many factors were associated with frailty, including age, cognitive impairment, depressive symptom, risk of malnutrition, ADL disability and poor self-perception of health.
Guidelines for future research	34	As a relatively neglected elderly group, evidence for the prevalence of frailty and its risk factors of rural elderly are important to the government to formulate policies related to aging.
Disclosure of funding source	35	No funding.

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Prevalence and risk factors of frailty among people in rural areas: a systematic review and meta-analysis

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Secondary Subject Heading:	Public health, Epidemiology, Global health
Keywords:	EPIDEMIOLOGY, PUBLIC HEALTH, Risk management < HEALTH SERVICES ADMINISTRATION & MANAGEMENT

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Prevalence and risk factors of frailty among people in rural areas: a systematic review and meta-analysis

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ABSTRACT

Objective Older people in rural areas are possibly more frail due to the limited medical resources and lower socioeconomic status. Given the negative healthy outcomes caused by frailty, knowing the epidemiology of frailty in rural areas is of great importance. We tried to synthesize the existing evidences for the prevalence and risk factors of frailty in rural areas.

Design A systematic review and meta-analysis.

Data sources PubMed, Embase, MEDLINE, Cochrane Library, Web of Science and Scopus were used to identify the articles from inception to 30 April, 2019.

Eligibility criteria Observational studies providing cross-sectional data on the prevalence of frailty in rural elderly were extracted.

Data extraction and synthesis Two independent investigators selected studies, extracted data and assessed the methodological quality of included studies. The pool prevalence of frailty was calculated by the random effects model and the odds ratio (OR) and 95% CI were used to calculated the risk factors.

Results The literature search yielded 2210 articles, of which 23 met the study criteria and were included in this analysis. The pooled prevalence of frailty and pre-frailty were 18% (95% CI 15% to 21%, I^2=98.5%, p＜0.001) and 50% (95% CI 45% to 56%, I^2=98.4%, p ＜ 0.001), respectively. The pooled frailty prevalence was 15% for the Fried Phenotype, 18% for the Frail Index, and 23% for other criteria. The pooled prevalence of frailty was 17% for males and 26% for females. The pooled prevalence of frailty was 17% in developing countries and 23% in developed countries. Age, cognitive impairment, depressive symptom, risk of malnutrition, ADL disability, and poor self-perception of health were associated with frailty. There was no publication bias.

Conclusions Frailty influences almost one in five older people in rural areas, and increasing age, cognitive impairment, depressive symptom, risk of malnutrition, ADL disability and poor self-perception of health were all risk factors for frailty. We should be cautious about the research results due to the heterogeneity between studies.

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Strengths and limitations of this study

(1) In this study, we conducted a comprehensive systematic literature search.

(2) This study focused on the frailty of older people in rural areas that are easily neglected.

(3) The data are mainly from Asia and North and South America, and lack of data from Europe.

(4) The main limitation is the heterogeneity among studies of this meta-analysis.

(5) Stratified meta-analyses cannot explain all sources of heterogeneity.

INTRODUCTION

An aging population is a common phenomenon experienced by all countries in the world,[1] which brings considerable challenges for the planning and delivery of healthcare services internationally. The most problematic manifestation of population aging is frailty.[2] Frailty is an age-related medical syndrome, which is characterized by diminished strength, endurance, and reduced physiologic function that increases an individual's vulnerability for developing increased dependency and/or death.[3] Frailty has been proved to be associated with negative healthy outcomes, such as hospitalization, institutionalization/dependency and premature mortalities.[4] Ahmad found that the transition towards greater frailty states were more likely (22.9%) than transition toward lesser frailty states (19.9%) during a 12-months follow-up through the survey of 2324 elderly and it proved that frailty was potential reversible and this reversibility made it a cornerstone to delay frailty progression.[5] Pre-frailty is an intermediate state between frailty and robustness, which can predispose to the development of frailty and other adverse health and social care outcomes and it is necessary to identify the state of pre-frailty.[6]

With the continuous expansion of the study population, more and more researches begin to pay attention to the frail elderly in rural areas. From high-income countries to sub Saharan Africa, under the guidance of city oriented development path, the gap between urban and rural areas continues to expand.[7] The elderly person in the rural

context, in addition to the peculiarities of the aging process,[8] shows a series of environment related characteristics, such as limited educational resources, limited accessibility of healthcare services and resources, lower socioeconomic status, and all of these potential factors may cause the emergence or advancement of frailty.[9 10] Many countries have explored the prevalence of frailty in rural elderly and the results are quite different, for example the prevalence of frailty is 43.4% in Brazil,[8] 15.2% in Sri Lanka,[11] 38.8% in India,[12] 18.8% in France[13] and even in the same country, the prevalence of frailty varies greatly due to differences in assessment tools, research environments and so on. A recent meta-analysis on prevalence of frailty including 62 countries across the world reported the prevalence of frailty and pre-frailty were 12% and 46% respectively using the tool of Fried phenotype (FP).[14] However, the included people were from cities, towns and villages, and the data of rural areas is absent.

Hence, it's necessary to synthesize the existing body of evidence and offer a perspective on the prevalence of frailty in rural areas, which are critically important to developing strategies and disease prevention. The objectives of this systematic review were to synthesize the pooled prevalence and risk factors of frailty in rural areas through conducting a meta-analysis.

METHODS

Protocol

This review was conducted according to the MOOSE statement which was for the meta-analysis of observational studies in epidemiology.[15]

Search strategy

We conducted a comprehensive search of the literature using six electronic databases. PubMed, Embase, MEDLINE, Cochrane Library, Web of Science and Scopus were searched from their dates of inception through 30 April 2020 with languages restricted to English. A combination of title/keyword/abstract and Medical Subject Headings (Mesh) were used and the following key words were used: "frailty", "frail", "frail*", "rural", "village", "countryside" and "rural*". Reference lists of relevant literatures were hand-searched for addition studies. The search strategy in PubMed is available in online supplemental table 1.

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Eligibility criteria

Condition

We considered studies reporting the prevalence (or enough data to compute this estimate) of frailty or pre-frailty. The measurement of frailty or pre-frailty was based on the assessment tools used. For example, the Fried phenotype contained 5 components (weight loss, exhaustion, weakness, slowness and low physical activity) and subjects who were positive for three or more components were defined as frailty, those with 1 or 2 component(s) were defined as pre-frailty.[16] In the studies where frailty or pre-frailty were not defined using the FP, we considered the definition used by authors.

Context

We considered studies conducted in people living in the rural areas. We found that there is no unified definition of rural areas by reviewing the literature. Some studies think that rural areas are the places with the relatively backward economic development,[10] while others think that rural areas are far away from cities.[5] Therefore, we will include studies as long as it clearly points out that it conducts in rural contexts.

Population

We considered studies conducted in older adults (⩾60 years). Studies that limited the study population to specific population such as lung cancer, kidney disease, and so on were excluded.

Study design

Study designs were observational providing cross-sectional data on the prevalence of frailty or pre-frailty. Review, editorial, comment, and conference abstract were excluded.

Data extraction

Two investigators extracted data independently and disagreements were resolved by discussion. Data were recorded in the Microsoft Excel file format from each eligible study including the following variables: first author, year and country of publication, study design, sample size, mean or median age of the population, proportion of

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female, assessment tools of frailty, prevalence of frailty and pre-frailty, and risk factors mentioned. We asked for the missing data by sending emails to the corresponding author of the included articles.

Methodological quality assessment

All eligible articles were subjected to a quality assessment by two investigators independently through the criterion proposed by Loney et al which was designed to critically assess studies about the prevalence.[17] Each study was assigned a summative quality score ranging from 0 to 8, which was obtained by scoring a point for each of: random sample or whole population, unbiased sampling frame (i.e. census data), adequate sample size (>300 subjects), measures were the standard, outcomes measured by unbiased assessors, adequate response rate (70%), confidence intervals, subgroup analysis, and study subjects described. Only studies that scored 3 or above were deemed eligible for inclusion and studies which met more than four criteria were thought to have adequate quality.[18] A third author resolved quality disagreements.

Statistical analysis

Heterogeneity among studies was tested using Chi-square test and the I^2 statistic. Studies were considered to have significant heterogeneity if the *p* value was less than 0.1. The I^2 statistic was used to assess the degree of heterogeneity, with I^2 values of 25%, 50%, and 75% being considered to indicate low, moderate, and high heterogeneity, respectively. Random-effects model was used to calculate the pooled prevalence of frailty if significant heterogeneity was present; otherwise a fixed-effects model was used. We tried to explore the potential reasons for heterogeneity by conducting stratified meta-analyses. In stratified meta-analyses, the literature data were divided into subgroups according to frailty criteria, gender, and level of development, and pooled estimates of frailty prevalence with 95% CIs were calculated. The prevalence of frailty and pre-frailty were extracted from all included studies in order to calculate the pooled prevalence and the odds ratios (ORs) and associated 95% CIs were extracted to assess the risk factors of frailty on rural elderly. Begg's test and Egger's test were used to assess publication bias.

Statistical analyses were conducted using Stata SE, version 12.0 (Stata Corp College

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Station, TX), and a p value of less than 0.05 was considered significant.

Patient and public involvement

There was no direct patient or public involvement in this study.

RESULTS

Study process

The systematic search of the six databases retrieved 2210 articles. After 1336 duplicate studies were excluded, a further 783 studies were excluded through screening the title and abstract. Of the 91 studies left, 68 were excluded from the full-text review due to non-data-based article (e.g. editorial, commentary, review, conference abstract) (n=15), same sample (n=11), non-English (n=7), no data of frailty in rural elderly (n=25), sample from a specific population (e.g. patients with cancer , disabled individuals) (n=6), one more form of frailty assessment (n=4). Finally, 23 published articles, [5 8-12 19-35] involving a total of 32478 older adults in rural areas, were eligible and included in the meta-analysis (Fig. 1). Of all the studies included, 17 studies have adequate quality (score＞4). The characteristics of studies were showed in Table 1. One of these studies included data from four countries.[35]

Prevalence of frailty and pre-frailty

The prevalence of frailty and pre-frailty in included studies range from 5.4% to 43.4%,[35 8] and from 23% to 67.8%,[23 22] respectively. Meta-analysis showed the pooled prevalence of frailty 18% (95% CI 15% to 21%, I^2=98.5%, p＜0.001) in rural elderly, with the pooled prevalence of pre-frailty of 50% (95% CI 45% to 56%, I^2=98.4%, p＜0.001) based on a random-effects model (Fig. 2 and Fig. 3). Heterogeneity statistics (Chi-square test=1644.51; p ＜ 0.001; I^2=98.5%) indicated significant variability among studies. According to the results of Begg'test (Pr=0.659＞0.05) and Egger's test (p=0.059＞0.05), there was no evidence of publication bias.

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Table 1 Characteristics of studies

First author(year)	Country	Sample size	Study design	mean or median age of the population	proportion of female	Frailty assessment	Prevalence (%)		Risk factors assessed	Quality Score
							frail	Pre-frail		
Llano2019(7)8	Brazil	820	Cross-sectional	—	56.10%	FP	43.4	37.1	income; education level; nutritional status，physical inactivity ,cognitive deficit, poor self-perceived health	6
Manrique-Espinoza2016(8)9	Mexico	558	Cross-sectional	—	47.5%	FP	8.6	52.9	ADL and IADL disability, depression symptom, educational level, self-reported health	6
Gu2019(9)10	China	4323	Cross-sectional	70.2±7.0	58.5%	FP	6.8	49.4	age, stroke history, vision decrease, anemia	6
Siriwardhana2019(10)11	Sri Lanka	746	Cross-sectional	68.0	53.2%	FP	15.2	48.5	age, longest-held occupation,	4
Dasgupta2019(11)12	India	165	Cross-sectional	66.99±6.5	67.9%	TFI	38.8	—	educational level，2 chronic diseases	4
Ahmad2018(18)5	Malaysia	2324	Based on cohort	—	62.1%	FP	9.4	57.9	—	6

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Table 1 Continued

First author(year)	Country	Sample size	Study design	mean or median age of the population	proportion of female	Frailty assessment	Prevalence (%)		Risk factors assessed	Quality Score
							frail	Pre-frail		
Huang2020(15)19	China	1014	Cross-sectional	78.7 ± 8.0	66.3%	FP	17.6	59.4	depressive symptoms; urinary incontinence; abnormal performance of TUG; malnutrition	5
Nguyen2019(16)20	Vietnam	512	Cross-sectional	—	69.9%	FP	21.7	65.6	age, marital status, occupation, comorbidities, cognitive impairment, history of fall in the last 12 months	5
Yoon2018(19)21	Korea	104	Cross-sectional	73.5±5.43	76.9%	FP	16.5	49.6	—	3
Li2018(20)22	China	3048	Cross-sectional	71.55±6.7	58.5%	FI	15.8	67.8	self-neglect, age, sex and alcohol drinking	5
Del,Brutto2016(21)23	Ecuador	311	Cross-sectional	71±8	56.9%	EFS	31.0	23.0	—	5
Dent2016(22)24	Australia	1501	Cross-sectional	75.9±7.9	54.9%	FI	25.0	—	health service use	6
Cakmur2015(26)25	Turkey	168	Cross-sectional	72.7±7.7	53.6%	FP	7.1	47.3	—	4

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Table 1 Continued

First author(year)	Country	Sample size	Study design	mean or median age of the population	proportion of female	Frailty assessment	Prevalence (%)		Risk factors assessed	Quality Score
							frail	Pre-frail		
Kim2019(23)26	Korea	808	Cross-sectional	74.58±6.26	59.0%	KYCL	35.5	—	perceived neighborhood walkability ,environmental pollution	6
Boulos2016(24)27	Lebanon	1120	Cross-sectional	75.7±7.1	59.4%	SOF	36.4	30.4	marital status, education level, Physical health status, Mental health status	6
Zhu2016(25)28	China	1478	Cross-sectional	75.3±3.9	53.0%	FP	12.0	42.9	high sensitivity C-reactive protein	6
Curcio2014(27)29	Colombia	1878	Based on cohort	70.9±7.4	52.2%	FP	12.2	53.0	age, gender, health status variables , functional covariate variables, psychosocial variables	4
Ocampo-Chaparro2019(28)30	Colombia	688	Cross-sectional	—	49.6%	FP	20.6	64.4	—	6

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Table 1 Continued

First author(year)	Country	Sample size	Study design	mean or median age of the population	proportion of female	Frailty assessment	Prevalence (%)		Risk factors assessed	Quality Score
							frail	Pre-frail		
Wu2017(29)31	China	3048	Based on cohort	—	—	FP	8.1	53.6	—	8
Abe2019(30)32	Japan	1576	Cross-sectional	—	55.5%	KYCL	20.8	—	—	8
Ma2018(31)33	China	2353	Cross-sectional	—	—	FI	12.9	—	—	7
Jung2016(32)34	Korea	382	Based on cohort	74.4±6.5	56.3%	FP	17.4	52.6	—	4
Llibre2018(33)35	Peru; Mexico; China; India	552; 1000; 1002; 999	Cross-sectional	74.2±7.3; 74.1±6.7; 72.4±6.0; 72.6±5.8	53.4%; 60.2%; 55.5%; 54.6%	the Modified FP	16.8; 15.7; 5.4; 15.4	—	—	7

Note: ADL=activity of daily living; IADL=instrumental activity of daily living; TUG=timed up-and-go; FP=the Fried Phenotype; FI=Frailty Index; TFI=Tilburg Frailty Indicator; EFS=the Edmonton Frail Scale; SOF=the Study of Osteoporotic Fractures index; KYCL=The Kaigo-Yobo checklist.

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Stratified prevalence of frailty according to frailty criteria, gender, level of development and region

The pooled estimates of frailty prevalence based on the FP, FI, and other criteria were 15%, 18%, and 32%, respectively. The estimated pooled prevalence of frailty was 26% in females and 17% in males. The estimated pooled prevalence of frailty was 17% in developing countries and 23% in developed countries. Results of subgroup analyses are shown in Table 2.

Table 2 Subgroup analyses by frailty criteria, gender, level of development and region

Subgroups	**Frailty**			
	Prevalence	**95%CI**	**I^2**	**P-value**
Frailty criteria				
FP	15%	12%-18%	97.7%	＜0.001
FI	18%	12%-24%	97.6%	＜0.001
other criteria	32%	24%-40%	96.4%	＜0.001
Gender				
Female	26%	20%-31%	98.4%	＜0.001
Male	17%	13%-22%	97.4%	＜0.001
Level of development				
Developing countries	17%	14%-20%	98.4%	＜0.001
Developed countries	23%	18%-29%	94.5%	＜0.001

Table 3 Pooled risk factors of frailty

No.	**Risk factors**	**OR**	**95%CI**	**I^2**	**P-value**
1	Age	1.05	1.03-1.08	0.0%	＜0.001
2	Cognitive impairment	1.97	1.41-2.54	0.0%	＜0.001
3	Depressive symptom	1.24	1.14-1.34	68.5%	＜0.001
4	Risk of malnutrition	2.49	1.51-3.48	62.0%	＜0.001
5	ADL disability	2.59	1.71-3.48	0.0%	＜0.001
6	Poor self-perception of health	2.42	1.39-3.45	0.0%	＜0.001

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Risk factors

We identified six potential risk factors including age, cognitive impairment, depressive symptom, risk of malnutrition, ADL disability, poor self-perception of health, which were associated with frailty in rural elderly for the pooled analysis (Table 3).

DISCUSSION

Based on 23 studies involving a total of 32478 participants living in rural areas, the estimated pooled prevalence of frailty and pre-frailty of older people in rural areas are 18% and 50%, respectively. However, this result has to be cautiously interpreted because of the substantial heterogeneity between studies. When assessing potential risk factors associated with frailty in rural elderly, six factors including increasing age, cognitive impairment, depressive symptom, risk of malnutrition, ADL disability and poor self-perception of health were associated with frailty.

The pooled prevalence of frailty in rural elderly from this meta-analysis was 18% (95% CI 15% to 21%), higher than the global estimate (10.7%; 95% CI 10.5% to 10.9%) which published in 2012.[36] Differences in frailty prevalence estimates between rural areas and the global average may be due to the characteristics of studies included. Most of the people included in the global study came from cities and towns, however the participants of our meta-analysis came from the rural areas. Frailty was more common among rural areas, which may be explained by these reasons: (1) lower socioeconomic status among residents living in rural communities, (2) limited accessibility of healthcare services and resources, (3) relatively unhealthy lifestyle and limited health care awareness.[19 20] Frailty has been proved to be associated with negative healthy outcomes,[4 37 38] and studies from Canada, China and Korea found that frail rural elderly have higher mortality than that in the city or town.[39-41] Besides, in the context of the COVID-19 pandemic, study found that frailty can predict disease outcomes better in a large population of patients with COVID-19 than age or comorbidity and it is also meaningful to know the frailty of older people in rural areas.[42] Hence, it's time to focus and improve the health care in rural areas, expect for

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improving the distribution of medical resources, strategies including support services, nutritional supplementation, exercise interventions and health education may also be useful to prevent and delay the occurrence and development of frailty.[34] The pooled prevalence of pre-frailty in this meta-analysis was found to be 50% (95% CI 45% to 56%). Pre-frailty is a transitional and potentially reversible risk state before the onset of frailty, and the fact that half of the rural elderly were pre-frail indicates that it is urgent to establish and implement early frailty prevention strategies among rural areas.

This analysis found that pooled prevalence of frailty varied based on the assessment tools used. In this meta-analysis, studies using the FP reported lower prevalence of frailty than those using the Frailty Index and other criteria, consistent with many other studies,[14 36] and this suggests that researchers around the world recognized the definition of FP. The benefits of the FP include its relative ease of application, and another benefit is the minimal amount of data required for its calculation, making it a potential tool for screening.[43] However, the FP has its roots in Canadian population surveys, and it is better to verify its applicability or to make an adjustment when used in other populations. The FI is another commonly used criterion, based on a deficit accumulation model,[44] and study found that the eFI (the electronic Frailty Index, eFI) overestimates the frailty status of community dwelling older people compared with the Clinical Frailty Scale which is a validated measure of frailty based on clinical presentation.[45] Hence, we need to consider carefully when using the FI and further work should also be conducted to identify a uniform scale that can be used to accurately identify frailty among populations.

Our stratified analysis of gender revealed that females were more likely to be frail than males among the older people in rural areas, which was consistent with previous reports.[14 36] Females have a longer life expectancy, and this survival effects may result in a greater accumulation of frailty-associated deficits over time.[46] Another possible explanation is that postmenopausal women have lower average amounts of lean body mass and muscle because of vitamin D deficiency,[47] and the relationship between frailty and sarcopenia has been confirmed in previous research.[48]

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The result of our level of development-stratified analysis revealed that older people in rural areas from developed countries were more likely to be frail than older people from developing countries, and this is not consistent with another meta-analysis which found that the prevalence of frailty appeared higher in community-dwelling older adults in upper middle-income countries compared with high-income countries.[49] This may be because older rural people in developed countries live longer than those in developing countries and further work should also be conducted to identify the differences in the incidence of frailty in countries at different levels of development.

The potential risk factors associated with frailty estimated in rural elderly were increasing age, cognitive impairment, and depressive symptom, risk of malnutrition, ADL disability and poor self-perception of health. Aging can be conceptualized as a deficit accumulation process of each physiological system, taking place in different individuals in different ways, with a variety of rates for different organ systems, making it necessary to find out prevalence and associated factors of different age groups of the elderly in rural areas.[50] Limited data made it difficult to conduct a subgroup of age in this meta-analysis, but evidence from rural elderly in Sri Lanka had found the prevalence of frailty has almost doubled with the increasing of age (group: 75-79 years vs ≥80 years; RR: 4.07 vs 7.02).[11] Park found that different age groups had different risk factors of frailty by analyzing the data of 22868 elderly people, so targeted interventions for frailty prevention can reduce the pressure in rural areas with limited medical resources.[51]

Cognitive impairment was associated with frailty in the rural elderly, consistent with results among the world that poor cognitive function is strongly closed associated with pre-frailty and frailty subgroups in older populations.[52] Many factors are involved in the relationship between cognitive function and frailty, such as neuropathology, cardiovascular disease, inflammation, hormonal changes, nutrition, social isolation, social vulnerability and so on.[53] As cognitive function contains several cognitive domains, it might be useful and necessary to determine which individual experienced cognitive deficits are most significantly associated with frailty scores.

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Depression was associated with frailty among older people in rural areas, and this is due to that frailty and depression share the same pathophysiological mechanism.[54] Besides, individuals with depressive symptoms usually have less physical activity, which will increase the probability of frailty.[55] Risk of malnutrition was another risk factors associated with frailty, as weight loss, reduced intake of calorie and specific dietary pattern may change the body composition and physical function, which can increase the risk of frailty.[55]

ADL disability means the decreased ability of self-care, affecting eating ability and physical activity, further causing malnutrition and muscle decline, eventually leading to frailty.[56] Self-perception of health has been identified as a highly sensitive variable in elderly health estimations and a valid predictor of morbidity, [9] and Jylha found self-appraisal has biological and cognitive underpinnings, existing as a predictor of frailty.[57] Actually, rural older persons are exposed to a greater inequitable and precarious socioeconomic and health conditions throughout their life courses, and as a result, perception of the elderly regarding adverse experiences may predispose them to frailty.

The strength of this study is to focus on the frailty of elderly in rural areas that are easily neglected. However, there are still several limitations. The main limitation is the heterogeneity among studies. Moreover, the included articles mainly from Asia, South and North America, lacking studies from Europe and Africa, perhaps due to the difference in number and distribution of rural population. Third, the quality of the studies included was moderate, and we should be careful with research results.

CONCLUSION

The urban-rural gap is the main problem in the distribution of medical resources in in various countries and there are a large number of older people in rural areas who are the key populations to achieve equalization of basic public health services. As a relatively neglected elderly group, a better insight of frailty and its risk factors will have wide-scale implications for health care policy and practice in rural dwelling populations. This meta-analysis found that the pooled prevalence of frailty of older people was 18% in rural areas and many factors were associated with frailty,

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including age, cognitive impairment, depressive symptom, risk of malnutrition, ADL disability and poor self-perception of health. Hence, it's time to focus the frailty of older people in rural areas. In order to prevent and reduce the frailty of the older adults in rural areas, we can try to establish a frailty screening mechanism for the elderly in rural areas, pay attention to the key groups, and establish nutrition, exercise, diet and other interventions.

Total number of words 3456

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Ethics approval Not applicable

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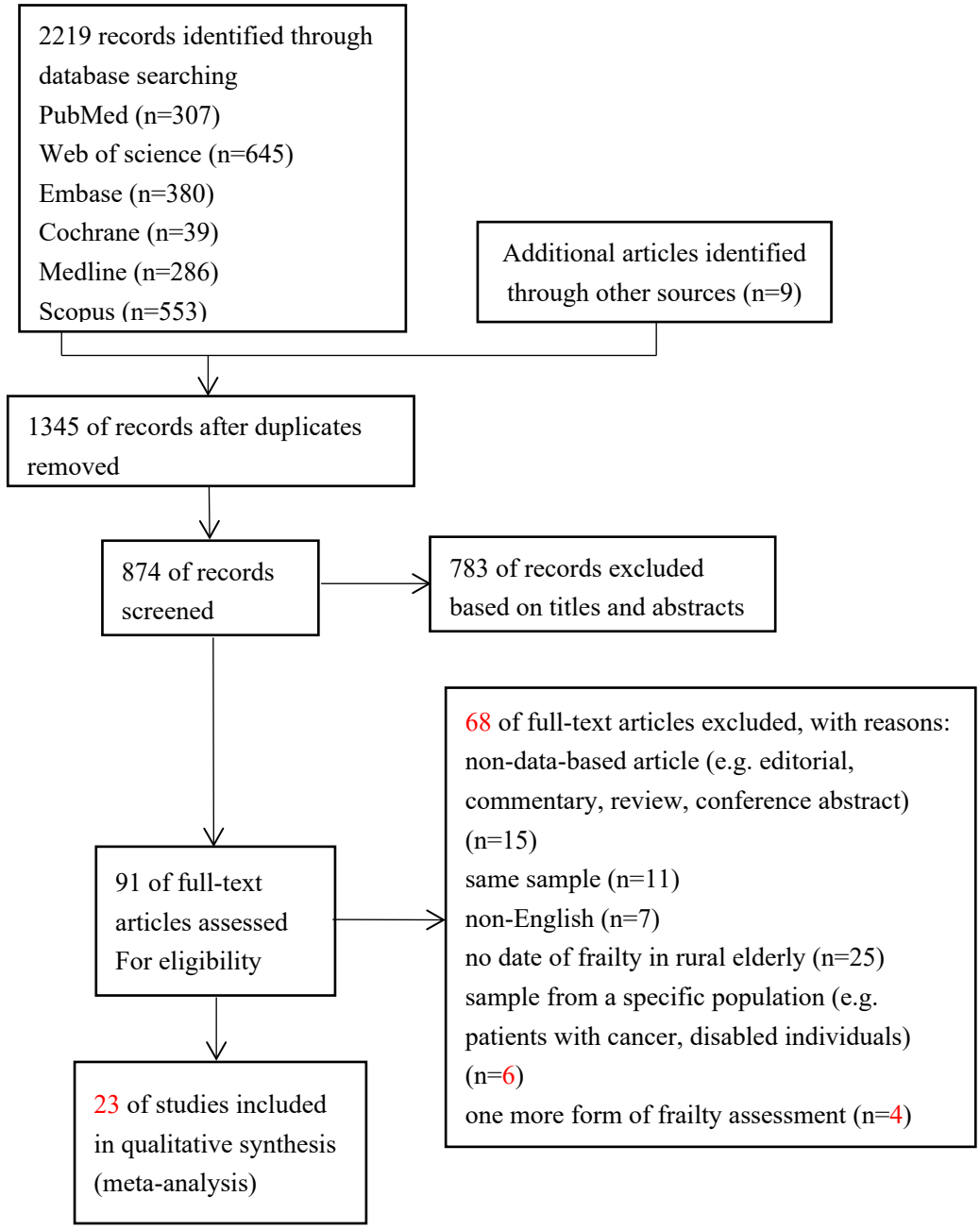
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Figure Legends

Fig 1. Flow chart of systematic review
Fig 2. Forest plot of prevalence of frailty
Fig 3. Forest plot of prevalence of pre-frailty

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Fig 1. Flow chart of systematic review



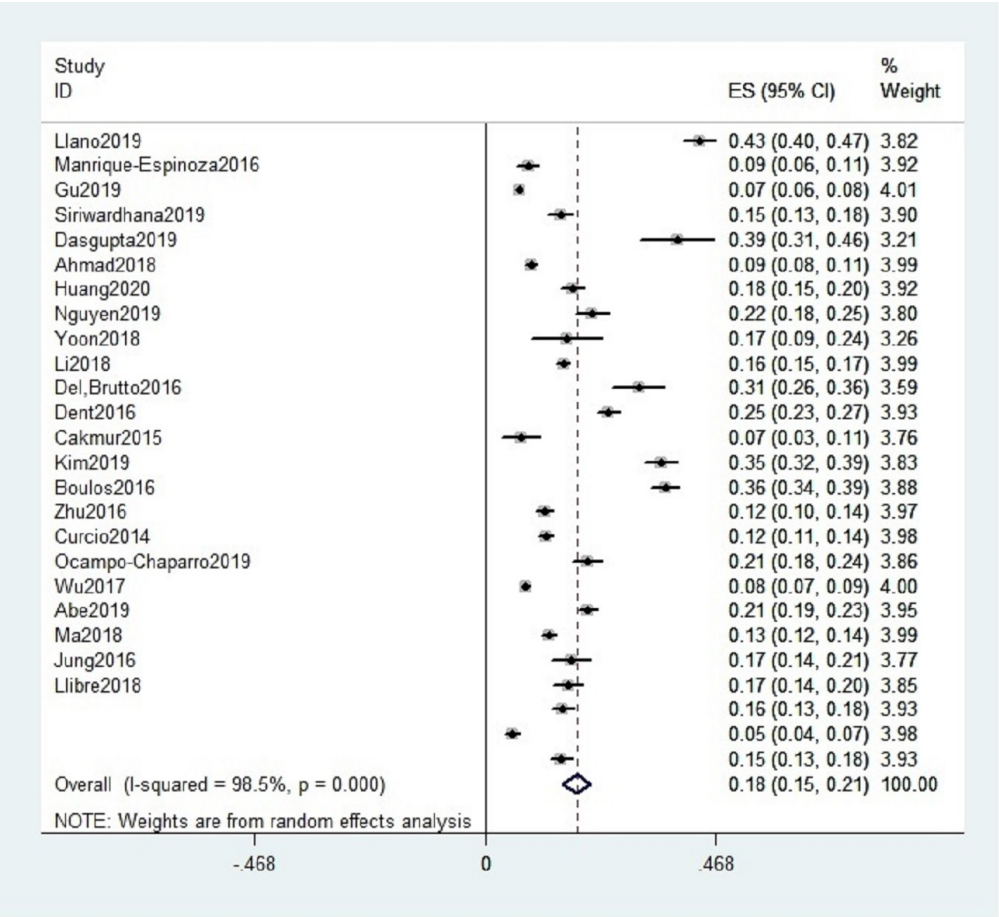


Fig 2. Forest plot of prevalence of frailty

109x100mm (300 x 300 DPI)

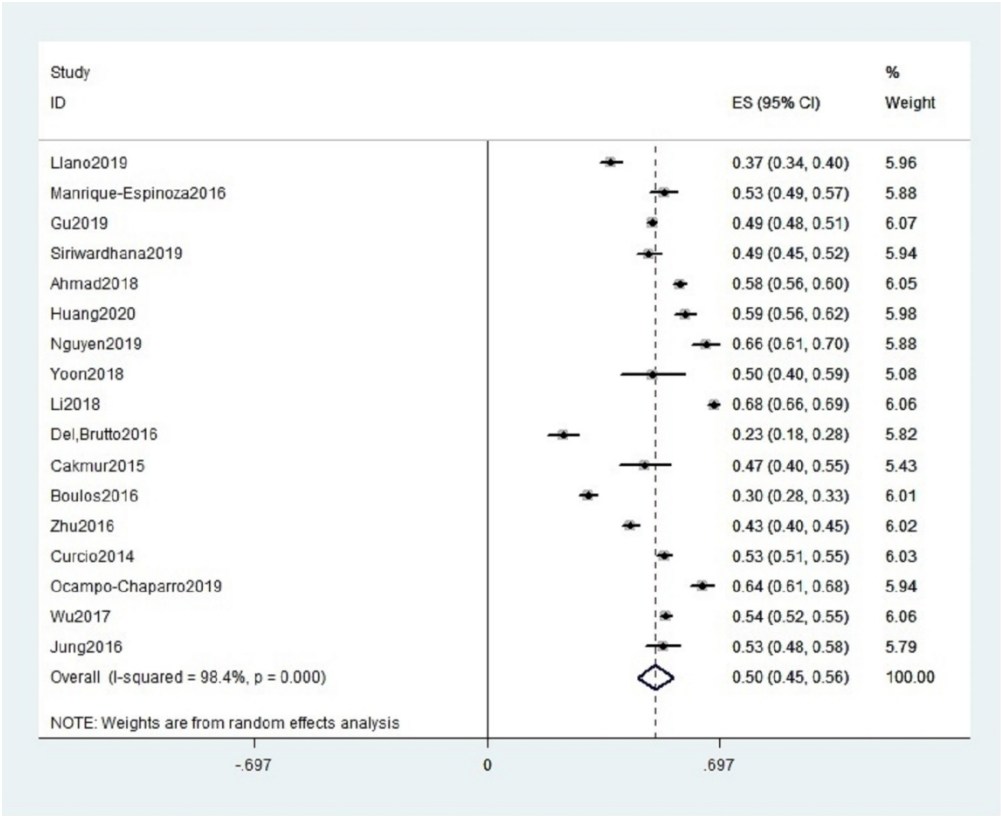


Fig 3. Forest plot of prevalence of pre-frailty

122x100mm (300 x 300 DPI)

Supplemental table 1

Database	Search strategy
PubMed	#1：(((frailty[Title/Abstract]) OR (frail[Title/Abstract])) OR (frail*[Title/Abstract])) OR (frailty[MeSH Terms]) Fliters: from the dates of inception to 30 April 2020 #2：(((rural[Title/Abstract]) OR (village[Title/Abstract]))OR (countryside[Title/Abstract])) OR (rural*[Title/Abstract]) Fliters: from the dates of inception to 30 April 2020 #3：#1 AND #2 Fliters: from the dates of inception to 30 April 2020

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MOOSE Checklist

Section	Item No.	Checklist Item
Background		
Problem definition	1	Synthesize the existing evidences for the prevalence and risk factors of frailty in rural areas.
Hypothesis statement	2	The elderly in rural areas has a high prevalence of frailty.
Description of study outcome(s)	3	The pooled prevalence of frailty and prefrailty were 18% and 50%, respectively. Age, cognitive impairment, depressive symptom, risk of malnutrition, ADL disability and poor self-perception of health were associated with frailty.
Type of exposure or intervention used	4	The prevalence of frailty.
Type of study designs used	5	Study designs were observational providing cross-sectional
Study population	6	Elderly in rural areas.
Search strategy		
Qualifications of searchers (eg, librarians and investigators)	7	This search was conducted by two postgraduates with relative training.
Search strategy , including time period included in the synthesis and keywords	8	It was shown in table 1.
Effort to include all available studies , including contact with authors	9	Reference lists of relevant literatures were hand-searched for addition studies in order to obtain all available studies.
Databases and registries searched	10	PubMed, Embase, Web of Science, Cochrane, Medline and Scopus were used to search.
Search software used, name and version , including special features used (eg, explosion)	11	It was EndNote x9.
Use of hand searching (eg, reference lists of obtained articles)	12	There were 9 articles through the hand searching.
List of citations located and those excluded, including justification	13	It was shown in Fig 1.
Method of addressing articles published in languages other than English	14	Not included.

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Continued

Section	Item No.	Checklist Item
Method of handling abstracts and unpublished studies	15	Not included.
Description of any contact with authors	16	There was no any connection with authors.
Methods		
Description of relevance or appropriateness of studies assembled for assessing the hypothesis to be tested	17	Studies included assembled for assessing the hypothesis to be tested.
Rationale for the selection and coding of data (eg, sound clinical principles or convenience)	18	Data were recorded including the following variables: first author, year and country of publication, sample size, assessment tools of frailty, prevalence of frailty and prefrailty, and risk factors mentioned.
Documentation of how data were classified and coded (eg, multiple raters, blinding, and interrater reliability)	19	All eligible articles were subjected to a quality assessment by two investigators independently.
Assessment of confounding (eg, comparability of cases and controls in studies where appropriate)	20	Not included.
Assessment of study quality, including blinding of quality assessors; stratification or regression on possible predictors of study results	21	The blinding was used for the assessment of study quality.
Assessment of heterogeneity	22	Heterogeneity statistics (Chi-square test=1651.51; p=0.000; I2=98.4%) indicated significant variability among studies.
Description of statistical methods (eg, complete description of fixed or random effects models, justification of whether the chosen models account for predictors of study results, dose-response models, or cumulative meta-analysis) in sufficient detail to be replicated	23	A random-effects model was used according to the results of heterogeneity (I^2=98.4%)

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Section	Item No.	Checklist Item
Provision of appropriate tables and graphics	24	Tables and graphics were shown in this article.
Results		
Graphic summarizing individual study estimates and overall estimate	25	It was shown in Fig 2.
Table giving descriptive information for each study included	26	It was shown in Table 2.
Results of sensitivity testing (eg, subgroup analysis)	27	It was shown in Table 3
Indication of statistical uncertainty of findings	28	Including the 95% CI and P-value.
Discussion		
Quantitative assessment of bias (eg, publication bias)	29	There was no evidence of publication bias according to the results of Begg' s test (Pr=0.650) and Egger ' s test (p=0.061).
Justification for exclusion (eg, exclusion of non- English-language citations)	30	In order to obtain the complete data and universal results.
Assessment of quality of included studies	31	The quality of included studies was in the level of moderate.
Conclusions		
Consideration of alternative explanations for observed results	32	The quality of original studies.
Generalization of the conclusions (ie, appropriate for the data presented and within the domain of the literature review)	33	The pooled prevalence of frailty of elderly was 18% in rural areas and many factors were associated with frailty, including age, cognitive impairment, depressive symptom, risk of malnutrition, ADL disability and poor self-perception of health.
Guidelines for future research	34	As a relatively neglected elderly group, evidence for the prevalence of frailty and its risk factors of rural elderly are important to the government to formulate policies related to aging.
Disclosure of funding source	35	No funding.

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